

MCQ

Assertion (A) : The polynomial $p(y) = y^2 + 4y + 3$ has two zeroes.

Reason (R) : A quadratic polynomial can have at most two zeroes.

- (a) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).
 (b) Both Assertion (A) and Reason (R) are true but Reason (R) is not the correct explanation of Assertion (A).
 (c) Assertion (A) is true, but Reason (R) is false.
 (d) Assertion (A) is false, but Reason (R) is true.  (2026) **Ap**

If the zeroes of a polynomial $p(x)$ are -3 and 8 , then $p(x)$ equals

- (a) $x^2 + 5x - 4$
 (b) $(x + 3)(-x + 8)$
 (c) $a(x^2 + 5x - 24)$
 (d) $x^2 - 24$  (2026) **Ap**

Zeroes of the polynomial $p(y) = 7y^2 - \frac{11}{3}y - \frac{2}{3}$ are :

- (a) $-\frac{2}{3}, -\frac{1}{7}$ (b) $-\frac{2}{7}, -\frac{1}{3}$
 (c) $\frac{2}{3}, \frac{1}{7}$ (d) $\frac{2}{3}, -\frac{1}{7}$  (2025) **Ap**

Zeroes of the polynomial $p(x) = x^2 - 3\sqrt{2}x + 4$ are :

- (a) $2, \sqrt{2}$ (b) $2\sqrt{2}, \sqrt{2}$
 (c) $4\sqrt{2}, -\sqrt{2}$ (d) $\sqrt{2}, 2$  (2025) **Ap**

The zeroes of the quadratic polynomial $2x^2 - 3x - 9$ are :

- (a) $3, \frac{-3}{2}$ (b) $-3, \frac{-3}{2}$
 (c) $-3, \frac{3}{2}$ (d) $3, \frac{3}{2}$  (2024) **Ap**

Assertion (A) : Degree of a zero polynomial is not defined.

Reason (R) : Degree of a non-zero constant polynomial is 0.

- (a) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).
 (b) Both Assertion (A) and Reason (R) are true but Reason (R) is not the correct explanation of Assertion (A).

(c) Assertion (A) is true, but Reason (R) is false.

(d) Assertion (A) is false, but Reason (R) is true.  (2024) **R**

7. The zeroes of the polynomial $3x^2 + 11x - 4$ are :

- (a) $\frac{1}{2}, -4$ (b) $\frac{1}{4}, -3$
 (c) $\frac{1}{3}, -4$ (d) $\frac{1}{3}, 4$ (2023C) **Ap**

8. The zeroes of the polynomial $p(x) = x^2 + 4x + 3$ are given by :

- (a) $1, 3$ (b) $-1, 3$
 (c) $1, -3$ (d) $-1, -3$ (2023) **Ap**

9. Which of the following is a quadratic polynomial with zeroes $\frac{5}{3}$ and 0 ?

- (a) $3x(3x - 5)$ (b) $3x(x - 5)$
 (c) $x^2 - \frac{5}{3}$ (d) $\frac{5}{3}x^2$ (2023)

10. If one of the zeroes of a quadratic polynomial $(k - 1)x^2 + kx + 1$ is -3 , then the value of k is

- (a) $\frac{4}{3}$ (b) $-\frac{4}{3}$ (c) $\frac{2}{3}$ (d) $-\frac{2}{3}$

(NCERT Exemplar, Term I, 2021-22) **Ev**

11. The degree of polynomial having zeroes -3 and 4 only is

- (a) 2 (b) 1
 (c) more than 3 (d) 3 (2020) **Ap**

12. If one of the zeroes of the quadratic polynomial $x^2 + 3x + k$ is 2 , then the value of k is

- (a) 10 (b) -10
 (c) -7 (d) -2 (2020)

VSA (1 mark)

13. If one zero of the quadratic polynomial $x^2 + 3x + k$ is 2 , then find the value of k . (2021C) **Ev**

SA I (2 marks)

14. Find the zeros of the polynomial

$$p(x) = x^2 + \frac{4}{3}x - \frac{4}{3}.$$

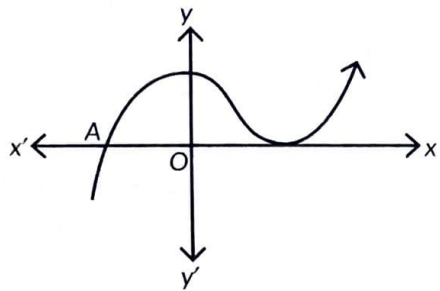
 (2025) **Ap**

2.2 Geometrical Meaning of the Zeroes of a Polynomial

[Page 11-18]

MCQ

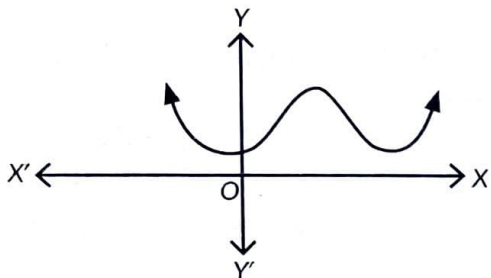
15. The graph of $y = f(x)$ is given. The number of distinct zeroes of $y = f(x)$ is



- (a) 0 (b) 1
(c) 2 (d) 3



6. The graph of $y = f(x)$ is given. The number of zeroes of $f(x)$ is:

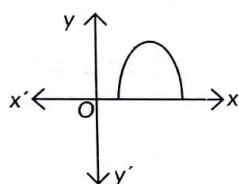


- (a) 0 (b) 1
(c) 2 (d) 4



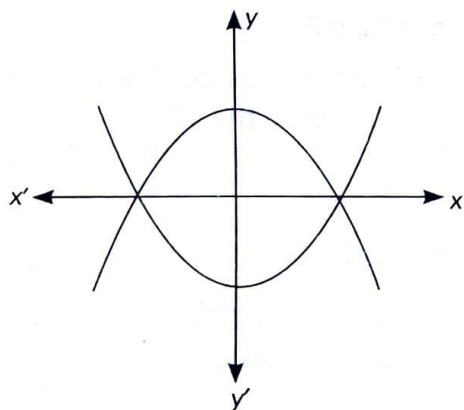
7. If the given figure shows the graph of polynomial $y = ax^2 + bx + c$, then:

- (a) $a < 0$
(b) $b^2 < 4ac$
(c) $c > 0$
(d) a and b are same sign



(2025C)

8. Two polynomials are shown in the graph below. The number of distinct zeroes of both the polynomials is:



- (a) 3 (b) 5
(c) 2 (d) 4



9. **Assertion (A)**: If the graph of a polynomial intersects the x -axis at exactly two points, then the number of zeroes of that polynomial is 2.

Reason (R): The number of zeroes of a polynomial is equal to the number of points where the graph of the polynomial intersects x -axis.

- (a) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).

(b) Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation for Assertion (A).

(c) Assertion (A) is true but Reason (R) is false.

(d) Assertion (A) is false but Reason (R) is true.

(2024C) R

20. **Assertion (A)**: If the graph of a polynomial touches x -axis at only one point, then the polynomial cannot be a quadratic polynomial.

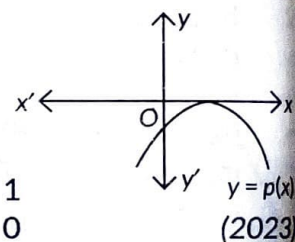
Reason (R): A polynomial of degree n ($n > 1$) can have at most n zeroes.

- (a) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).
(b) Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation for Assertion (A).
(c) Assertion (A) is true but Reason (R) is false.
(d) Assertion (A) is false but Reason (R) is true.

(2024) R

21. The graph of $y = p(x)$ is given, for a polynomial $p(x)$. The number of zeroes of $p(x)$ from the graph is

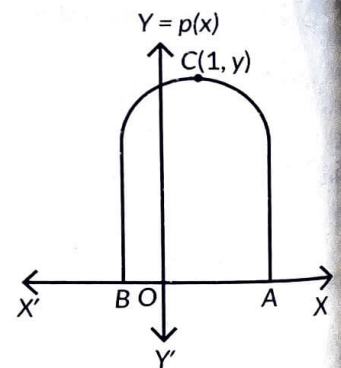
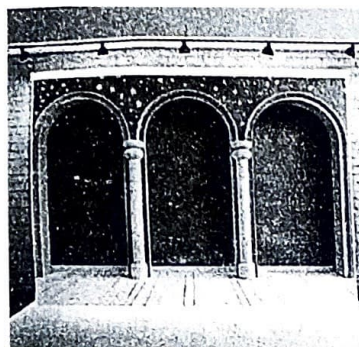
- (a) 3 (b) 1
(c) 2 (d) 0



(2023)

CBQ (4 marks)

22. **Case Study**: During a theatre drama, a backdrop of building arches was used. The shape of the curve shown below can be represented by the polynomial $p(x) = -x^2 + 2x + 8$, where x is the length (in feet) on stage level.



Based on the figure given above, answer the following questions:

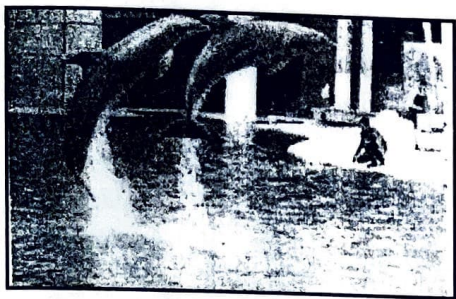
- (i) Determine the height of the arch.
(ii) (a) Find zeroes of the polynomial $p(x)$. Which points on the graph represent the zeroes?

OR

- (ii) (b) Find the span on the arch on the stage floor.
(iii) Write the coordinates of the point of intersection of the above curve with the y -axis.

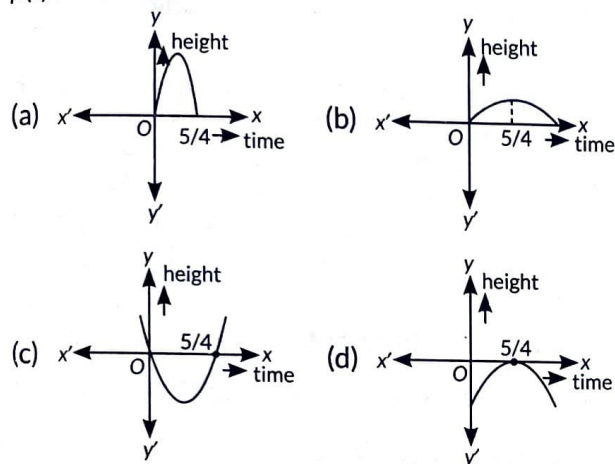
(2026)

23. **Case Study :** In a pool at an aquarium, a dolphin jumps out of the water travelling at 20 cm per second. Its height above water level after t seconds is given by $h = 20t - 16t^2$.



Based on the above, answer the following questions :

- (i) Find zeroes of polynomial $p(t) = 20t - 16t^2$.
 (ii) Which of the following types of graph represents $p(t)$?



- (iii) (a) What would be the value of h at $t = \frac{3}{2}$? Interpret the result.

OR

- (iii) (b) How much distance has the dolphin covered before hitting the water level again? (2023) CFQ

HWT 2.3 Relationship between Zeroes and Coefficients of a Polynomial

[Page 18-23]

MCQ

24. The sum and product of zeroes of a quadratic polynomial $p(x)$ are $-\frac{1}{3}$ and 2 respectively. The polynomial $p(x)$ is
 (a) $3x^2 - x + 6$ (b) $x^2 + \frac{1}{3}x - 2$
 (c) $3x^2 - x + 2$ (d) $-3x^2 - x - 6$ (2026) (Ap)
25. If α and β are two zeroes of a polynomial $f(x) = px^2 - 2x + 3p$ and $\alpha + \beta = \alpha\beta$, then value of p is
 (a) $-\frac{2}{3}$ (b) $\frac{2}{3}$
 (c) $\frac{1}{3}$ (d) $-\frac{1}{3}$ (2026) (Ap)

26. If sum and product of zeroes of a polynomial are (-3) and (-2) respectively, then a polynomial is
 (a) $x^2 - 3x - 2$ (b) $-x^2 - 3x + 2$
 (c) $-x^2 + 3x - 2$ (d) $x^2 + 3x + 2$ (2026)

27. If α and β are the zeroes of the polynomial $p(x) = x^2 - ax - b$, then the value of $(\alpha + \beta + \alpha\beta)$ is equal to:

- (a) $a + b$ (b) $-a - b$
 (c) $a - b$ (d) $-a + b$ (2025) (Ev)

28. If α and β are zeroes of the polynomial $p(x) = kx^2 - 30x + 45k$ and $\alpha + \beta = \alpha\beta$, then the value of ' k ' is:

- (a) $-\frac{2}{3}$ (b) $-\frac{3}{2}$
 (c) $\frac{3}{2}$ (d) $\frac{2}{3}$ (2025, 2024) (Ev)

29. If α and β are the zeroes of polynomial $3x^2 + 6x + k$ such that $\alpha + \beta + \alpha\beta = -\frac{2}{3}$, then the value of k is:

- (a) -8 (b) 8
 (c) -4 (d) 4 (2025) (Ev)

30. If the zeroes of the polynomial $ax^2 + bx + \frac{2a}{b}$ are reciprocal of each other, then the value of b is

- (a) 2 (b) $\frac{1}{2}$
 (c) -2 (d) $-\frac{1}{2}$ (2025) (Ev)

31. If the sum of zeroes of the polynomial

$p(x) = 2x^2 - k\sqrt{2}x + 1$ is $\sqrt{2}$, then value of k is:

- (a) $\sqrt{2}$ (b) 2
 (c) $2\sqrt{2}$ (d) $1/2$ (2024) (Ev)

32. The zeroes of a polynomial $x^2 + px + q$ are twice the zeroes of the polynomial $4x^2 - 5x - 6$. The value of p is:

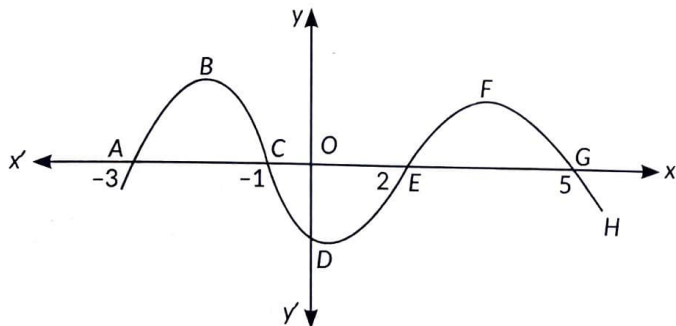
- (a) $-\frac{5}{2}$ (b) $\frac{5}{2}$
 (c) -5 (d) 10 (2024) (Ev)

33. If the sum and the product of zeroes of a quadratic polynomial are $2\sqrt{3}$ and 3 respectively, then a quadratic polynomial is:

- (a) $x^2 + 2\sqrt{3}x - 3$
 (b) $(x - \sqrt{3})^2$
 (c) $x^2 - 2\sqrt{3}x - 3$
 (d) $x^2 + 2\sqrt{3}x + 3$ (2024) (Ap)

34. If one zero of the polynomial $6x^2 + 37x - (k - 2)$ is reciprocal of the other, then what is the value of k ?
 (a) -4 (b) -6
 (c) 6 (d) 4 (2023) (Ev)
35. If α and β are the zeroes of the quadratic polynomial $p(x) = x^2 - ax - b$, then the value of $\alpha^2 + \beta^2$ is:
 (a) $a^2 - 2b$ (b) $a^2 + 2b$
 (c) $b^2 - 2a$ (d) $b^2 + 2a$ (2023) (Ev)
36. The number of polynomials having zeroes -3 and 5 is:
 (a) only one
 (b) infinite
 (c) exactly two
 (d) atmost two (2023) (An)
37. If α, β are the zeroes of a polynomial $p(x) = x^2 + x - 1$, then $\frac{1}{\alpha} + \frac{1}{\beta}$ equals to
 (a) 1 (b) 2 (c) -1 (d) $-\frac{1}{2}$ (2023)
38. If α, β are zeroes of the polynomial $x^2 - 1$, then value of $(\alpha + \beta)$ is
 (a) 2 (b) 1
 (c) -1 (d) 0 (2023)
39. If α, β are the zeroes of the polynomial $p(x) = 4x^2 - 3x - 7$, then $\left(\frac{1}{\alpha} + \frac{1}{\beta}\right)$ is equal to
 (a) $\frac{7}{3}$ (b) $-\frac{7}{3}$ (c) $\frac{3}{7}$ (d) $-\frac{3}{7}$ (2023)

Case study for Q. No. 40 to 44: A car moves on a highway. The path it traces is given below:



Based on the above information, attempt any 4 questions out of 5.

40. What is the shape of the curve EFG?
 (a) Parabola (b) Ellipse
 (c) Straight line (d) Circle

(Term I, 2021-22) (An)

41. If the curve ABC is represented by the polynomial $-(x^2 + 4x + 3)$, then its zeroes are
 (a) 1 and -3 (b) -1 and 3
 (c) 1 and 3 (d) -1 and -3 (Term I, 2021-22)
42. If the path traced by the car has zeroes at -1 and 2 , then it is given by
 (a) $x^2 + x + 2$ (b) $x^2 - x + 2$
 (c) $x^2 - x - 2$ (d) $x^2 + x - 2$ (Term I, 2021-22) (An)
43. The number of zeroes of the polynomial representing the whole curve, is
 (a) 4 (b) 3 (c) 2 (d) 1 (Term I, 2021-22) (An)
44. The distance between C and G is
 (a) 4 units (b) 6 units
 (c) 8 units (d) 7 units (Term I, 2021-22)
45. The quadratic polynomial, the sum of whose zeroes is -5 and their product is 6 , is
 (a) $x^2 + 5x + 6$ (b) $x^2 - 5x + 6$
 (c) $x^2 - 5x - 6$ (d) $-x^2 + 5x + 6$ (Term I, 2021-22, 2020)

VSA (1 mark)

46. If α and β are the zeroes of the quadratic polynomial $f(x) = x^2 - x - 4$, find the value of $\frac{1}{\alpha} + \frac{1}{\beta} - \alpha\beta$. (2021 C)
47. If α, β are zeroes of the polynomial $2x^2 - 5x - 4$, then $\frac{1}{\alpha} + \frac{1}{\beta} =$ _____. (2020 C) (U)
48. If α, β are zeroes of the polynomial $-3x^2 + x - 5$, then the value of $\frac{1}{\alpha} + \frac{1}{\beta}$ is _____. (2020 C)
49. Form a quadratic polynomial, the sum and product of whose zeroes are -3 and 2 respectively. (2020) (Ap)

SA I (2 marks)

50. Find the value of p , for which one zero of the quadratic polynomial $px^2 - 14x + 8$ is 6 times the other. (2026) (Ev)
51. If α, β are the zeroes of the quadratic polynomial $px^2 + qx + r$, then find the value of $\alpha^3\beta + \beta^3\alpha$. (2026) (Ev)
52. If α, β are the zeroes of the polynomial $p(x) = x^2 - 3x - 1$, then find the value of $\frac{1}{\alpha} + \frac{1}{\beta}$. (2026) (Ev)
53. If the sum of the zeroes of the polynomial $p(x) = (p + 1)x^2 + (2p + 3)x + (3p + 4)$ is -1 , then find the value of ' p '. (2025) (R)
54. If α and β are zeroes of the polynomial $p(x) = x^2 - 2x - 1$, then find the value of $\frac{1}{2\alpha} + \frac{1}{2\beta} + 3\alpha\beta$. (2025) (U)

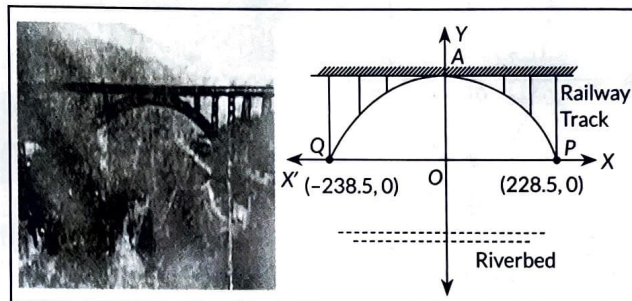
55. If ' α ' and ' β ' are the zeroes of the polynomial $p(y) = y^2 - 5y + 3$, then find the value of $\alpha^4\beta^3 + \alpha^3\beta^4$. (2025) 
56. If the zeroes of the polynomial $x^2 + ax + b$ are in the ratio 3 : 4, then prove that $12a^2 = 49b$. (2025) 
57. If α, β are zeroes of the polynomial $p(x) = 5x^2 - 6x + 1$, then find the value of $\alpha + \beta + \alpha\beta$. (2024) 
58. If one zero of the polynomial $p(x) = 6x^2 + 37x - (k - 2)$ is reciprocal of the other, then find the value of k . (2023)

SA II (3 marks)

59. α, β are the zeroes of the quadratic polynomial $p(x) = x^2 - 4x + k$, such that $\alpha - \beta = 8$. Find the value of k . (2025C) 
60. Find the zeroes of the polynomial $q(x) = 8x^2 - 2x - 3$. Hence, find a polynomial whose zeroes are 2 less than the zeroes of $q(x)$. (2025)
61. Find the zeroes of the polynomial $p(x) = 3x^2 - 4x - 4$. Hence, write a polynomial whose each of the zeroes is 2 more than zeroes of $p(x)$. (2025) 
62. Find a quadratic polynomial whose sum of the zeroes is 8 and difference of the zeroes is 2. (2024C)
63. Find the zeroes of the quadratic polynomial $x^2 - 15$ and verify the relationship between the zeroes and the coefficients of the polynomial. (2024) 
64. Find a quadratic polynomial whose zeroes are reciprocals of the zeroes of the polynomial $f(x) = ax^2 + bx + c, a \neq 0, c \neq 0$. (2020) 
65. Find the value of k such that the polynomial $x^2 - (k + 6)x + 2(2k - 1)$ has sum of its zeroes equal to half of their product. (Delhi 2019)
66. Find the zeroes of the quadratic polynomial $7y^2 - \frac{11}{3}y - \frac{2}{3}$ and verify the relationship between the zeroes and the coefficients. (2019)
67. Find the quadratic polynomial, sum and product of whose zeroes are -1 and -20 respectively. Also, find the zeroes of the polynomial so obtained. (2019)
68. If α and β are zeroes of $4x^2 - x - 4$, find quadratic polynomial whose zeroes are $\frac{1}{2\alpha}$ and $\frac{1}{2\beta}$. (Board Term I, 2017) 
69. If α and β are the zeroes of $p(x) = 6x^2 - 7x + 2$. Find the quadratic polynomial whose zeroes are $\frac{1}{\alpha}$ & $\frac{1}{\beta}$. (Board Term I, 2017)

CBQ (4 marks)

70. **Case Study :** An arch of a railway bridge, built on Chenab riverbed, is shown in the above diagram. It is a parabolic arch connecting two hills at P and Q. If the parabolic curve is represented by the polynomial $p(x) = -0.0025x^2 - 0.025x + 136$.



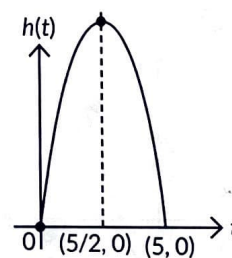
Observe the diagram and based on above information, answer the following questions :

- Write the co-ordinates of point A.
- Find the span of the arch.
- (a) Write the zeroes of the polynomial using diagram and verify the relationship between sum of zeroes and polynomials.

OR

- (b) Find the values of $p(x)$ at $x = 100$ and $x = -100$. Are they same? (2026) 

71. **Case Study :** A ball is thrown in the air so that t seconds after it is thrown, its height h metre above its starting point is given by the polynomial $h = 25t - 5t^2$.



Observe the graph of the polynomial and answer the following questions :

- Write zeroes of the given polynomial.
 - Find the maximum height achieved by ball.
 - (a) After throwing upward, how much time did the ball take to reach to the height of 30 m?
- OR
- (b) Find the two different values of t when the height of the ball was 20 m. (2024) 

72. **Case Study :** While playing in a garden, Samaira saw a honeycomb and asked her mother what is that. Her mother replied that it's a honeycomb made by

1. (a) : The polynomial $p(y) = y^2 + 4y + 3$ is a quadratic polynomial.

Thus, the polynomial has exactly two distinct zeroes. So, assertion (A) is true.

Hence, both assertion (A) and reason (R) are true and reason (R) is the correct explanation of the assertion (A).

2. (b) : Let α and β are the zeroes of polynomial $p(x)$.

Sum of zeroes, $\alpha + \beta = -3 + 8 = 5$

Product of zeroes, $\alpha\beta = (-3)(8) = -24$

$\therefore$ Required polynomial = $K[x^2 - (\alpha + \beta)x + \alpha\beta]$
 $= K[x^2 - 5x - 24] = -x^2 + 5x + 24$ [Put $K = -1$]
 $= (x + 3)(-x + 8)$

3. (d) : The given polynomial is $p(y) = 7y^2 - \frac{11y}{3} - \frac{2}{3}$

$$\Rightarrow p(y) = \frac{21y^2 - 11y - 2}{3} = \frac{(7y+1)(3y-2)}{3}$$

For zeroes of the polynomial, put $p(y) = 0$

$$\Rightarrow \frac{(7y+1)(3y-2)}{3} = 0 \Rightarrow y = \frac{-1}{7}, \frac{2}{3}$$

$$= x(x - 2\sqrt{2}) - \sqrt{2}(x - 2\sqrt{2}) = (x - 2\sqrt{2})(x - \sqrt{2})$$

For zeroes of the polynomial, put $p(x) = 0$

$$\Rightarrow (x - 2\sqrt{2})(x - \sqrt{2}) = 0$$

$\therefore x = 2\sqrt{2}, \sqrt{2}$ are zeroes of $p(x)$.

5. (a) : We have, $p(x) = 2x^2 - 3x - 9$

$$\Rightarrow p(x) = 2x^2 - 6x + 3x - 9 \\ = 2x(x - 3) + 3(x - 3) = (x - 3)(2x + 3)$$

For zeros of the polynomial $p(x)$,

Put $p(x) = 0$

$$\Rightarrow (x - 3)(2x + 3) = 0$$

$$\Rightarrow (x - 3) = 0 \text{ or } (2x + 3) = 0 \Rightarrow x = 3, -3/2$$

6. (b)

7. (c) : The given polynomial is $3x^2 + 11x - 4$
 $= 3x^2 + 12x - x - 4 = 3x(x + 4) - 1(x + 4) = (3x - 1)(x + 4)$

So, the value of $3x^2 + 11x - 4$ is zero when

$$x = 1/3 \text{ or } x = -4$$

$\therefore 1/3$ and -4 are zeroes of the given polynomial.

8. (d) : $p(x) = x^2 + 4x + 3 = x^2 + 3x + x + 3$
 $= x(x + 3) + 1(x + 3) = (x + 3)(x + 1)$

The value of $p(x)$ is zero, when $x + 3 = 0$ or $x + 1 = 0$
i.e., when $x = -3$ or $x = -1$

So, zeroes of polynomial $p(x)$ are -3 and -1 .

9. (a) : Consider option (a)

$$3x(3x - 5) = 0$$

$$\Rightarrow 3x = 0 \text{ or } 3x - 5 = 0 \Rightarrow x = 0 \text{ or } x = \frac{5}{3}$$

10. (a) : Given, -3 is a zero of quadratic polynomial $(k - 1)x^2 + kx + 1$.

$$\therefore (k - 1)(-3)^2 + k(-3) + 1 = 0$$

$$\Rightarrow 9k - 9 - 3k + 1 = 0 \Rightarrow 6k - 8 = 0 \Rightarrow k = 8/6 \Rightarrow k = \frac{4}{3}$$

11. (a) : Since, the polynomial has two zeroes only. So, the degree of the polynomial is 2.

Commonly Made Mistake

Don't confuse the highest power of x with the number of terms in the polynomial.

The degree of the polynomial is the highest power of the variable in a polynomial not the number of terms.

12. (b) : Given, 2 is a zero of the polynomial

$$p(x) = x^2 + 3x + k$$

$$\therefore p(2) = 0 \Rightarrow (2)^2 + 3(2) + k = 0 \Rightarrow 4 + 6 + k = 0$$

$$\Rightarrow 10 + k = 0 \Rightarrow k = -10$$

13. Given, polynomial is $f(x) = x^2 + 3x + k$

Since, 2 is zero of the polynomial $f(x)$.

$$\therefore f(2) = 0$$

$$\Rightarrow f(2) = (2)^2 + 3 \times 2 + k$$

$$\Rightarrow 4 + 6 + k = 0 \Rightarrow k = -10$$

14. Given polynomial is $p(x) = x^2 + \frac{4}{3}x - \frac{4}{3}$

$$\Rightarrow p(x) = \frac{3x^2 + 4x - 4}{3} = \frac{3x^2 + 6x - 2x - 4}{3}$$

$$= \frac{3x(x+2) - 2(x+2)}{3} = \frac{(3x-2)(x+2)}{3} \quad (1/2)$$

For zeros of the polynomial $p(x)$, put $p(x) = 0$ (1/2)

$$\Rightarrow \frac{(3x-2)(x+2)}{3} = 0$$

$$\Rightarrow 3x - 2 = 0 \Rightarrow x = \frac{2}{3} \text{ or } x + 2 = 0 \Rightarrow x = -2$$

$$\therefore x = -2 \text{ or } x = \frac{2}{3} \text{ are zeroes of the given polynomial.} \quad (1)$$

15. (c) : Here, $y = f(x)$ touches the x -axis at one point and cuts the x -axis at one point.

So, number of zeroes is two.

16. (a) : In the given graph, the curve intersects the x -axis at zero points. Therefore, the number of zeroes is 0.

17. (a) : $y = ax^2 + bx + c$ represents a parabola opening downward.

$$\therefore a < 0.$$

18. (c) : Since, both the polynomials cuts the x -axis at two distinct point. So, the number of distinct zeroes of both the polynomials is 2.

19. (a) : Its obvious, number of zeroes depends on the number of intersection made by graph of polynomial or x -axis.

20. (d) : Assertion is false, as if the graph of a polynomial touches x -axis at only one point, then the polynomial can be a quadratic polynomial having real and equal roots. Clearly, Reason is true.

21. (b) : Here, $y = p(x)$ touches the x -axis at one point. So, number of zeros is one.

22. (i) Height of the arch will be $p(1)$

$$\text{Now, } p(1) = -(1)^2 + 2(1) + 8 = 9 \text{ feet} \quad (1)$$

$$(ii) (a) p(x) = -x^2 + 2x + 8 = -x^2 + 4x - 2x + 8$$

$$= -x(x - 4) - 2(x - 4) = (x - 4)(-x - 2) \quad (1)$$

$$\Rightarrow p(x) = 0$$

$$\Rightarrow x = 4 \text{ and } x = -2 \quad \dots (i)$$

Hence, the zeroes of the polynomial are $x = -2$ and $x = 4$

In the graph, B represent $(-2, 0)$ and A represent $(4, 0)$. (1)

OR

(ii) (b) Since point A and B represent the zeroes, then span = $AB = 4 - (-2) = 6$ feet (2)

(iii) When curve cuts y -axis, then $x = 0$, i.e. $y = p(0)$

$$\text{Now, } p(0) = 8$$

Hence, required coordinates are $(0, 8)$ (1)

23. (i) We have, $p(t) = 20t - 16t^2 = 4t(5 - 4t)$

To find zeroes of polynomial, we have

$$4t = 0 \text{ or } 5 - 4t = 0$$

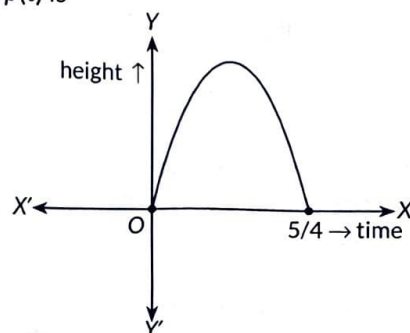
$$\Rightarrow t = 0 \text{ or } t = 5/4$$

$\therefore$ Zeroes of given polynomial are 0 and $5/4$. (1)

(ii) (a) : The values of the polynomial should be zero at $t = 0$ and $t = \frac{5}{4}$. So, graph given by option (b) and (d) can't be possible.

Now, let's check the value of $p(t)$ between 0 and $\frac{5}{4}$
Put $t = 1$, then we get

$$p(1) = 20(1) - 16(1)^2 = 20 - 16 = 4, \text{ which is positive. So, graph of } p(t) \text{ is}$$



(iii) (a) As $h = p(t) = 20t - 16t^2$

$$\text{At } t = \frac{3}{2}, h = 20\left(\frac{3}{2}\right) - 16\left(\frac{3}{2}\right)^2 = 30 - 36 = -6 \quad (1\frac{1}{2})$$

$\therefore h = -6$ cm, which shows the dolphin is 6 cm below water level. (1/2)

OR

(iii) (b) As the dolphin hits the water level in $\frac{5}{4}$ sec. when travelling with 20 cm per second. (1)

$$\therefore \text{Distance covered} = \text{speed} \times \text{time} = 20 \times \frac{5}{4} = 25 \text{ cm.} \quad (1)$$

24. (d) : Let α and β are the zeroes of a quadratic polynomial $p(x)$. Then

$$\text{Sum of zeroes} = \alpha + \beta = -\frac{1}{3}$$

$$\text{Product of zeroes} = \alpha\beta = 2$$

$\therefore$ Required polynomial, $p(x) = k[x^2 - (\alpha + \beta)x + \alpha\beta]$ where k is any real number.

$$= k\left[x^2 - \left(-\frac{1}{3}\right)x + 2\right] = k\left[x^2 + \frac{1}{3}x + 2\right] \quad \dots (i)$$

Put $k = -3$ in (i), we get

$$p(x) = -3x^2 - x - 6$$

25. (b) : Given polynomial, $f(x) = px^2 - 2x + 3p$

α, β are the roots of $px^2 - 2x + 3p = 0$

$$\text{So, } \alpha + \beta = \frac{2}{p} \text{ and } \alpha\beta = 3$$

Given, $\alpha + \beta = \alpha\beta$

$$\Rightarrow \frac{2}{p} = 3 \Rightarrow p = \frac{2}{3}$$

Concept Applied

For the quadratic polynomial $ax^2 + bx + c$

$$\text{sum of roots} = -\frac{b}{a} \text{ and product of roots} = \frac{c}{a}$$

26. (b) : Let α and β are the zeroes of polynomial

Sum of zeroes, $\alpha + \beta = -3$

Product of zeroes, $\alpha\beta = -2$

$$\therefore \text{Required polynomial} = k[x^2 - (\alpha + \beta)x + \alpha\beta] \\ = k[x^2 + 3x - 2] = -x^2 - 3x + 2 \quad [\text{Put } k = -1]$$

27. (c) : Given polynomial is

$$p(x) = x^2 - ax - b, \text{ and } \alpha \text{ and } \beta \text{ are zeroes of } p(x)$$

$\therefore$ Sum of zeroes $= \alpha + \beta = a$

Product of zeroes $= \alpha\beta = -b$

Now, $\alpha + \beta + \alpha\beta = a - b$

Concept Applied

If α and β are the zeroes of quadratic polynomial $p(x) = ax^2 + bx + c$, then $\alpha + \beta = -b/a$, $\alpha\beta = c/a$.

28. (d) : Given, α and β are zeroes of the polynomial $p(x) = kx^2 - 30x + 45k$

Here, $a = k$, $b = -30$, $c = 45k$

$$\text{Sum of zeroes} = \alpha + \beta = \frac{-b}{a} = \frac{30}{k}$$

$$\text{Product of zeroes} = \alpha\beta = \frac{c}{a} = \frac{45k}{k} = 45$$

$$\therefore \alpha + \beta = \alpha\beta$$

$$\Rightarrow \frac{30}{k} = 45 \Rightarrow k = \frac{30}{45} = \frac{2}{3}$$

29. (d) : Compare $3x^2 + 6x + k$ with $ax^2 + bx + c$, we get $a = 3$, $b = 6$ and $c = k$

$$\therefore \alpha + \beta + \alpha\beta = \left(-\frac{b}{a}\right) + \frac{c}{a} \Rightarrow \frac{-2}{3} = \frac{-6}{3} + \frac{k}{3}$$

$$\Rightarrow \frac{k}{3} = \frac{-2}{3} + \frac{6}{3} \Rightarrow \frac{k}{3} = \frac{4}{3} \Rightarrow k = 4$$

30. (a) : Let α and β be the zeroes of the given polynomial $ax^2 + bx + \frac{2a}{b}$.

$$\text{Given } \beta = \frac{1}{\alpha}$$

$$\Rightarrow \text{Product of zeroes} = \alpha \cdot \beta = \frac{\text{Constant term}}{\text{Coefficient of } x^2}$$

$$\Rightarrow \alpha \cdot \frac{1}{\alpha} = \frac{2a/b}{a} \Rightarrow 1 = \frac{2a}{b \times a} \Rightarrow 1 = \frac{2}{b} \Rightarrow b = 2$$

31. (b) : Given, sum of zeroes of the given polynomial is $\sqrt{2}$

$$\Rightarrow \frac{(-k\sqrt{2})}{2} = \sqrt{2} \Rightarrow k = \frac{2\sqrt{2}}{\sqrt{2}} = 2$$

32. (a) : Let the zeroes of the polynomial $x^2 + px + q$ be α and β . Then, $\alpha + \beta = -p$, $\alpha\beta = q$... (i)

Also, for the zeroes of the polynomial $p(x) = 4x^2 - 5x - 6$, put $p(x) = 0$

$$\Rightarrow 4x^2 - 5x - 6 = 0 \Rightarrow 4x^2 - 8x + 3x - 6 = 0$$

$$\Rightarrow 4x(x - 2) + 3(x - 2) = 0 \Rightarrow (x - 2)(4x + 3) = 0$$

$$\Rightarrow x = 2 \text{ or } x = -\frac{3}{4} \text{ are the zeroes of } p(x) = 0.$$

But twice the zeroes of $p(x)$ = zeroes of $x^2 + px + q$

$$\Rightarrow \alpha = 4 \text{ and } \beta = -\frac{3}{2} \therefore \alpha + \beta = 4 - \frac{3}{2} = \frac{5}{2}$$

$$\therefore \text{The value of } p = -\frac{5}{2}.$$

33. (b) : Sum of zeroes $= 2\sqrt{3}$

Product of zeroes $= 3$

$$\therefore \text{Required quadratic polynomial is } k[x^2 - 2\sqrt{3}x + 3] \\ = k[x^2 - 2 \times \sqrt{3} \times x + (\sqrt{3})^2] = [k(x - \sqrt{3})^2]$$

Thus, one of the polynomial which satisfy the given condition is $(x - \sqrt{3})^2$.

34. (a) : Let α and $\frac{1}{\alpha}$ be two zeroes of quadratic polynomial $6x^2 + 37x - (k - 2)$ then $\alpha \times \frac{1}{\alpha} = \frac{-(k-2)}{6}$

$$\Rightarrow -k + 2 = 6 \Rightarrow -k = 4 \Rightarrow k = -4$$

35. (b) : Sum of zeroes of quadratic polynomial

$$= \frac{-\text{coefficient of } x}{\text{coefficient of } x^2} \text{ and product of zeroes} = \frac{\text{constant term}}{\text{coefficient of } x^2}$$

$$\therefore \alpha + \beta = a, \alpha\beta = -b$$

Now, consider $(\alpha + \beta)^2 = \alpha^2 + \beta^2 + 2\alpha\beta$

$$\Rightarrow a^2 = \alpha^2 + \beta^2 - 2b$$

$$\Rightarrow \alpha^2 + \beta^2 = a^2 + 2b$$

36. (b) : The quadratic polynomial having zeroes α and β can be obtained by $k(x^2 - (\alpha + \beta)x + \alpha\beta)$, where k is any real number.

$\therefore k[x^2 - (-3 + 5)x + (-3)(5)] = k[x^2 - 2x - 15]$ are the polynomials having zeroes -3 and 5 .

So, there exists infinite number of such polynomials.

37. (a) : We have, $p(x) = x^2 + x - 1$, $\alpha + \beta = -1$ and $\alpha\beta = -1$

$$\text{Now, } \frac{1}{\alpha} + \frac{1}{\beta} = \frac{\alpha + \beta}{\alpha\beta} = \frac{-1}{-1} = 1$$

38. (d) : Since, α, β are the zeroes of polynomial $x^2 - 1$
or $x^2 + 0x - 1$

$\therefore$ Sum of zeroes, $(\alpha + \beta) = 0$

39. (d) : Since, α, β are the zeroes of polynomial
 $p(x) = 4x^2 - 3x - 7$

$\therefore$ Sum of zeroes, $(\alpha + \beta) = \frac{3}{4}$

and product of zeroes $(\alpha\beta) = \frac{-7}{4}$

$$\text{Now, } \frac{1}{\alpha} + \frac{1}{\beta} = \frac{\alpha + \beta}{\alpha\beta} = \frac{\frac{3}{4}}{\frac{-7}{4}} = \frac{-3}{7}$$

40. (a) : Shape of curve EFG is parabola.

Expert Tips

➤ A parabola is a curve where any point on the parabola is at equal distance from a fixed line and a fixed point (not on the line).

41. (d) : Given polynomial is $-(x^2 + 4x + 3)$.

Corresponding equation is $x^2 + 4x + 3 = 0$

$$\Rightarrow x^2 + x + 3x + 3 = 0 \Rightarrow (x + 1)(x + 3) = 0 \Rightarrow x = -1, -3$$

42. (c) : The polynomial having zeroes α, β is
 $k[x^2 - (\alpha + \beta)x + \alpha\beta]$, where k is real.

Here $\alpha = -1$ and $\beta = 2$

$$\therefore \text{Required polynomial} = k[x^2 - (-1 + 2)x + (-1) \times (2)]$$

$$= x^2 - x - 2 \quad (\text{for } k = 1)$$

Alternative Method

Since, -1 and 2 are zeroes, then polynomial can also be written as

$$p(x) = (x + 1)(x - 2)$$

$$\Rightarrow p(x) = x^2 + x - 2x - 2 \Rightarrow p(x) = x^2 - x - 2$$

43. (a) : Given curve cuts the x -axis at four distinct points. So, number of zeroes will be 4.

Commonly Made Mistake

➤ Remember the difference between x -axis and y -axis.

44. (b) : The distance between point C and G is 6 units.

45. (a) : Let α, β be the zeroes of required polynomial $p(x)$.
Given, $\alpha + \beta = -5$ and $\alpha\beta = 6$

$$\therefore p(x) = k[x^2 - (-5)x + 6] = k[x^2 + 5x + 6]$$

Thus, one of the polynomial which satisfy the given condition is $x^2 + 5x + 6$.

Concept Applied

➤ Quadratic polynomial $= k[x^2 - (\text{sum of zeroes})x + \text{product of zeroes}]$

46. Given, α and β are the zeros of the polynomial
 $f(x) = x^2 - x - 4$.

$$\therefore \alpha + \beta = 1 \text{ and } \alpha\beta = -4$$

$$\text{Now, } \frac{1}{\alpha} + \frac{1}{\beta} - \alpha\beta = \frac{\alpha + \beta}{\alpha\beta} - \alpha\beta = \frac{1}{-4} - (-4) = \frac{-1}{4} + 4 = \frac{15}{4}$$

Concept Applied

➤ For the quadratic polynomial $ax^2 + bx + c$ sum of roots
 $= \frac{-b}{a}$ and product of roots $= \frac{c}{a}$

47. Given, α and β are zeros of $2x^2 - 5x - 4$

$$\therefore \alpha + \beta = 5/2 \text{ and } \alpha\beta = -2$$

$$\text{Now, } \frac{1}{\alpha} + \frac{1}{\beta} = \frac{\alpha + \beta}{\alpha\beta} = \frac{5/2}{-2} = -5/4$$

48. Given, α and β are zeros of $-3x^2 + x - 5$

$$\therefore \alpha + \beta = 1/3 \text{ and } \alpha\beta = 5/3$$

$$\text{Now, } \frac{1}{\alpha} + \frac{1}{\beta} = \frac{\alpha + \beta}{\alpha\beta} = \frac{1/3}{5/3} = 1/5$$

49. Let α, β be the zeroes of required polynomial $p(x)$.
Given, $\alpha + \beta = -3$ and $\alpha\beta = 2$

$$\therefore p(x) = k[x^2 - (-3)x + 2] = k[x^2 + 3x + 2]$$

For $k = 1$, $p(x) = x^2 + 3x + 2$

Hence, one of the polynomial which satisfy the given condition is $x^2 + 3x + 2$.

50. Given equation is $px^2 - 14x + 8 = 0$

Let roots of the equation be α, β .

$$\text{Then, } \beta = 6\alpha \Rightarrow 6\alpha - \beta = 0 \quad \dots(i)$$

$$\text{Also, sum of roots} = \alpha + \beta = -\left(\frac{-14}{p}\right) = \frac{14}{p} \quad \dots(ii)$$

$$\text{and product of roots} = \alpha\beta = \frac{8}{p} \quad \dots(iii) \quad (1)$$

On solving (i) and (ii), we get

$$\alpha = \frac{2}{p} \text{ and } \beta = \frac{12}{p}$$

Putting these values in (iii), we get

$$\left(\frac{2}{p}\right) \times \left(\frac{12}{p}\right) = \frac{8}{p}$$

$$\Rightarrow 8p^2 = 24p \Rightarrow 8p(p - 3) = 0$$

$$\Rightarrow p = 3$$

$$(\because p \neq 0) \quad (1)$$

51. Given polynomial is $px^2 + qx + r$.

On comparing the polynomial with $ax^2 + bx + c$, we get
 $a = p, b = q$ and $c = r$

$$\therefore \alpha + \beta = \frac{-b}{a} = \frac{-q}{p}, \alpha\beta = \frac{c}{a} = \frac{r}{p}$$

$$\therefore \alpha^3\beta + \beta^3\alpha = \alpha\beta(\alpha^2 + \beta^2) = \alpha\beta[(\alpha + \beta)^2 - 2\alpha\beta] \quad (1)$$

$$= \frac{r}{p} \left[\left(\frac{-q}{p}\right)^2 - \frac{2r}{p} \right] = \frac{r}{p} \left[\frac{q^2 - 2rp}{p^2} \right] = \frac{r(q^2 - 2rp)}{p^3} \quad (1)$$

52. For $p(x) = x^2 - 3x - 1$, we have $a = 1, b = -3$ and $c = -1$

$$\therefore \alpha + \beta = \frac{-3}{1} = -3 \text{ and } \alpha\beta = \frac{-1}{1} = -1 \quad (1/2)$$

$$\therefore \frac{1}{\alpha} + \frac{1}{\beta} = \frac{\alpha + \beta}{\alpha\beta} = \frac{-3}{-1} = 3 \quad (1/2)$$

Given polynomial is

$$p(x) = (p+1)x^2 + (2p+3)x + (3p+4)$$

Let α and β are zeroes of given polynomial

$$\therefore \text{Sum of zeroes} = \alpha + \beta = \frac{-b}{a} = \frac{-(2p+3)}{(p+1)} \quad (1)$$

$$\Rightarrow -1 = \frac{-(2p+3)}{(p+1)} \quad (\because \text{sum of zeroes} = -1)$$

$$\Rightarrow p+1 = 2p+3 \Rightarrow p = -2$$

Hence, the value of p is -2 . (1)

54. The given polynomial is $p(x) = x^2 - 2x - 1$;

if α and β are zeroes of given polynomial

$$\therefore \text{Sum of zeroes} = \alpha + \beta = \frac{-b}{a} = 2 \quad \dots(i) \quad (1/2)$$

$$\text{and product of zeroes} = \alpha\beta = \frac{c}{a} = -1 \quad \dots(ii) \quad (1/2)$$

$$\begin{aligned} \text{Now, } \frac{1}{2\alpha} + \frac{1}{2\beta} + 3\alpha\beta &= \frac{\beta + \alpha + 6(\alpha\beta)(\alpha\beta)}{2\alpha\beta} \\ &= \frac{2 + 6(-1)(-1)}{2(-1)} \quad [\because \text{From (i) and (ii)}] \\ &= -4 \quad (1) \end{aligned}$$

55. We have $p(y) = y^2 - 5y + 3$

Let α and β be zeroes of $p(y)$.

$$\text{Given, sum of zeroes} = \alpha + \beta = 5 \quad (1/2)$$

$$\text{Product of zeroes} = \alpha\beta = 3 \quad (1/2)$$

$$\begin{aligned} \alpha^4\beta^3 + \alpha^3\beta^4 &= \alpha^3\beta^3(\alpha + \beta) \\ &= (\alpha\beta)^3(\alpha + \beta) = (3)^3(5) = 27 \times 5 = 135 \quad (1) \end{aligned}$$

56. Let the zeroes of the polynomial $x^2 + ax + b$ be $3x$ and $4x$.

$$\therefore 3x + 4x = -a \Rightarrow a = -7x \text{ or } x = \frac{-a}{7} \quad \dots(i)$$

$$\text{And } (3x)(4x) = b \Rightarrow b = 12x^2 \quad (1)$$

$$\Rightarrow b = 12\left(\frac{-a}{7}\right)^2 \quad [\because \text{Using (i)}]$$

$$\Rightarrow 49b = 12a^2. \text{ Hence proved.} \quad (1)$$

57. Given, α and β are zeroes of the polynomial

$$p(x) = 5x^2 - 6x + 1.$$

Here, $a = 5$, $b = -6$ and $c = 1$

$$\text{Sum of roots} = \frac{-b}{a} \Rightarrow \alpha + \beta = \frac{-(-6)}{5} = \frac{6}{5} \quad (1/2)$$

$$\text{Product of roots} = \frac{c}{a} \Rightarrow \alpha\beta = \frac{1}{5} \quad (1/2)$$

$$\text{Now, } \alpha + \beta + \alpha\beta = \frac{6}{5} + \frac{1}{5} = \frac{6+1}{5} = \frac{7}{5} \quad (1)$$

58. Let one zero of the polynomial $p(x)$ be α , then the other zero be $1/\alpha$.

$$\text{For zeroes of polynomial } p(x) = 6x^2 + 37x - (k-2) \quad (1/2)$$

$$\therefore \text{Product of zeroes} = \alpha \times \frac{1}{\alpha} = \frac{-(k-2)}{6} \quad (1/2)$$

$$\Rightarrow \frac{-(k-2)}{6} = 1 \Rightarrow k-2 = -6$$

$$\Rightarrow k = -4 \quad (1)$$

59. Given, α and β are the zeroes of the polynomial

$$p(x) = x^2 - 4x + k.$$

$$\therefore \alpha + \beta = 4 \quad \dots(i)$$

$$\alpha\beta = k \quad \dots(ii)$$

$$\alpha - \beta = 8 \quad \dots(iii)$$

$$\text{Now, } (\alpha - \beta)^2 = (\alpha + \beta)^2 - 4\alpha\beta \quad (1)$$

$$\Rightarrow (8)^2 = (4)^2 - 4k \text{ (using (i), (ii) and (iii))}$$

$$\Rightarrow 64 = 16 - 4k \Rightarrow 4k = -48 \quad (1)$$

$$\Rightarrow k = \frac{-48}{4}$$

$$\Rightarrow k = -12 \quad (1)$$

60.

$$\begin{aligned} q(x) &= 8x^2 - 2x - 3 \\ 8x^2 - 2x - 3 \\ \text{Let zeroes of } q(x) \text{ be } \alpha \text{ and } \beta. \\ 8x^2 - 2x - 3 \\ 8x^2 - 6x + 4x - 3 \\ 2x(4x-3) + 1(4x-3) \\ (2x+1)(4x-3) \\ \text{Comparing with } k(x-\alpha)(x-\beta), \\ \alpha = -\frac{1}{2}, \beta = \frac{3}{4} \\ \text{Comparing } q(x) \text{ with } ax^2 + bx + c, \\ a=8, b=-2, c=-3 \\ \text{Now, } \alpha + \beta = -\frac{b}{a} \quad \alpha\beta = \frac{c}{a} \\ \alpha + \beta = -\left(-\frac{2}{8}\right) \quad \alpha\beta = \frac{-3}{8} \\ \alpha + \beta = \frac{1}{4} \\ \text{Now, let the required polynomial be } p(x) \\ \text{Then, zeroes of } p(x) \rightarrow \alpha - 2 \text{ and } \beta - 2 \\ \text{Sum of zeroes} \rightarrow \alpha - 2 + \beta - 2 = \alpha + \beta - 4 \\ = \frac{1}{4} - 4 \quad (\because \alpha + \beta = \frac{1}{4}) \\ = \frac{1-16}{4} \\ \text{Sum of zeroes} = -\frac{15}{4} \\ \text{Product of zeroes} \rightarrow (\alpha - 2)(\beta - 2) = \alpha\beta - 2\alpha - 2\beta + 4 \\ = (\alpha\beta) - 2(\alpha + \beta) + 4 \\ = \frac{-3}{8} - 2\left(\frac{1}{4}\right) + 4 \quad (\because \alpha\beta = \frac{-3}{8} \text{ and } \alpha + \beta = \frac{1}{4}) \\ = \frac{-3}{8} - \frac{1}{2} + 4 \\ = \frac{-3-4+32}{8} \\ = \frac{25}{8} \\ \text{Now, } p(x) = k[x^2 - (\alpha + \beta)x + \text{product of zeroes}] \\ p(x) = k[x^2 - \text{sum of zeroes}x + \text{product of zeroes}] \\ = k\left[x^2 - \left(-\frac{15}{4}\right)x + \frac{25}{8}\right] \text{ where } k \text{ is a constant} \\ = k\left[8x^2 + 30x + 25\right] \\ \text{Thus, } p(x) = k[8x^2 + 30x + 25] \\ \text{OR if } k = \frac{1}{8}, p(x) = 8x^2 + 30x + 25 \\ \text{Alternate} \rightarrow \alpha = -\frac{1}{2}, \beta = \frac{3}{4} \\ \text{Let required polynomial be } p(x) \\ \text{The zeroes of } p(x) = \alpha - 2 \text{ and } \beta - 2 \\ = -\frac{1}{2} - 2 \text{ and } \frac{3}{4} - 2 \\ = -\frac{5}{2} \text{ and } \frac{3-8}{4} \\ = -\frac{5}{2} \text{ and } -\frac{5}{4} \\ q(x) = k[x^2 - \text{sum of zeroes}x + \text{product of zeroes}] \end{aligned}$$

61. The polynomial is $p(x) = 3x^2 - 4x - 4$.

The zeroes are given by $p(x) = 0$

$$\Rightarrow 3x^2 - 4x - 4 = 0 \Rightarrow 3x^2 - 6x + 2x - 4 = 0$$

$$\Rightarrow 3x(x - 2) + 2(x - 2) = 0 \Rightarrow (x - 2)(3x + 2) = 0$$

$$\Rightarrow x - 2 = 0 \text{ or } 3x + 2 = 0 \Rightarrow x = 2 \text{ and } -\frac{2}{3}$$

Thus, zeroes of new polynomial are

$$x = 2 + 2 \text{ and } x = -\frac{2}{3} + 2 \Rightarrow x = 4 \text{ and } x = \frac{4}{3}$$

Hence, new polynomial is

$$r(x) = a(x - 4)\left(x - \frac{4}{3}\right), \text{ where 'a' is any constant.}$$

$$= a\left(x^2 - 4x - \frac{4x}{3} + \frac{16}{3}\right) = a\left(x^2 - \frac{16}{3}x + \frac{16}{3}\right)$$

Taking $a = 3$, $r(x) = 3x^2 - 16x + 16$

Thus, $r(x) = 3x^2 - 16x + 16$ is the new polynomial with zeroes $x = 4$ and $x = \frac{4}{3}$.

62. Let α, β be the zeroes of a quadratic polynomial.

$$\therefore \alpha + \beta = 8$$

$$\text{and } \alpha - \beta = 2$$

[Given]

On solving these equations we get $\alpha = 5$ and $\beta = 3$

$\therefore$ sum of zeroes = 8 and product of zeroes = 15

Required quadratic polynomial

$$= x^2 - (\text{sum of zeroes})x + \text{product of zeroes}$$

$$= x^2 - 8x + 15$$

63. Let the given quadratic polynomial is $p(x) = x^2 - 15$

For zeroes, $p(x) = 0$

$$\Rightarrow x^2 = 15 \Rightarrow x = \pm\sqrt{15}$$

$$\therefore \text{Zeroes are } \sqrt{15}, -\sqrt{15}.$$

Verification

$$\text{Sum of zeroes} = \sqrt{15} + (-\sqrt{15}) = \frac{0}{1} = \frac{-(\text{coefficient of } x)}{\text{coefficient of } x^2}$$

and product of zeroes

$$= \sqrt{15}(-\sqrt{15}) = -15 = \frac{-15}{1} = \frac{\text{constant term}}{\text{coefficient of } x^2}$$

64. Let α, β are the zeroes of the polynomial $ax^2 + bx + c$.

$$\therefore \alpha + \beta = -\frac{b}{a} \text{ and } \alpha\beta = \frac{c}{a}$$

Now we have to find quadratic polynomial whose zeroes are $\frac{1}{\alpha}$ and $\frac{1}{\beta}$.

$$\text{Now, } \frac{1}{\alpha} + \frac{1}{\beta} = \frac{\alpha + \beta}{\alpha\beta} = \frac{-b/a}{c/a} = \frac{-b}{c} \text{ and } \left(\frac{1}{\alpha}\right)\left(\frac{1}{\beta}\right) = \frac{1}{\alpha\beta} = \frac{1}{c/a} = \frac{a}{c}$$

Thus, the required polynomial is

$$p(x) = k\left(x^2 - \left(-\frac{b}{c}\right)x + \frac{a}{c}\right) = \frac{k}{c}(cx^2 + bx + a)$$

For $k = c$, $p(x) = cx^2 + bx + a$

Expert Tips

➤ A quadratic polynomial is a second-degree polynomial, where the value of the highest degree is equal to 2.

65. The given polynomial is $f(x) = x^2 - (k + 6)x + 2(2k - 1)$

Now, sum of zeroes = $\frac{k+6}{1}$ and

$$\text{product of zeroes} = \frac{2(2k-1)}{1}$$

According to question,

$$\text{Sum of zeroes} = \frac{1}{2}(\text{Product of zeroes}) \Rightarrow k + 6 = \frac{1}{2} \times 2(2k - 1)$$

$$\Rightarrow k + 6 = 2k - 1 \Rightarrow k = 7$$

$$66. \text{ Let } f(y) = 7y^2 - \frac{11}{3}y - \frac{2}{3} \text{ or } \frac{21y^2 - 11y - 2}{3}$$

$$\Rightarrow f(y) = \frac{(3y-2)(7y+1)}{3}$$

Zeros of polynomial is given by $f(y) = 0$

$$\Rightarrow (3y - 2)(7y + 1) = 0 \Rightarrow \text{Either } y = \frac{2}{3} \text{ or } y = -\frac{1}{7}$$

Hence, $\frac{-1}{7}$ and $\frac{2}{3}$ are zeros of $f(y)$.

Verification

$$\text{Sum of zeros} = \frac{-1}{7} + \frac{2}{3} = \frac{-3+14}{21} = \frac{11}{21} = -\frac{(\text{Coefficient of } y)}{(\text{Coefficient of } y^2)}$$

$$\text{and product of zeros} = \left(\frac{-1}{7}\right)\left(\frac{2}{3}\right) = \frac{-2}{21} = \frac{\text{Constant term}}{\text{Coefficient of } y^2}$$

Hence, the relationship between zeros and coefficients of the polynomial is verified.

67. Let the zeros of the quadratic polynomial be α and β .

Given, $\alpha + \beta = -1$ and $\alpha \cdot \beta = -20$

Then, the quadratic polynomial will be

$$f(x) = x^2 - (\alpha + \beta)x + \alpha\beta$$

$$\Rightarrow f(x) = x^2 - (-1)x + (-20) \Rightarrow f(x) = x^2 + x - 20$$

To find the zeros of the polynomial, we put $f(x) = 0$

$$\Rightarrow x^2 + x - 20 = 0 \Rightarrow x^2 + 5x - 4x - 20 = 0$$

$$\Rightarrow x(x + 5) - 4(x + 5) = 0 \Rightarrow x = 4, -5$$

$\therefore x = 4$ and -5 are the zeroes of the polynomial so obtained.

68. Given, α and β are the zeroes of the polynomial $4x^2 - x - 4$.

$$\therefore \alpha + \beta = \frac{1}{4} \text{ and } \alpha\beta = \frac{-4}{4} = -1$$

Let S and P denote respectively the sum and product of the zeroes of the polynomial whose zeroes are $\frac{1}{2\alpha}$ and $\frac{1}{2\beta}$.

$$\text{Then, } S = \frac{1}{2\alpha} + \frac{1}{2\beta} = \frac{\alpha + \beta}{2\alpha\beta} = \frac{\frac{1}{4}}{2(-1)} = \frac{-1}{8}$$